

Yichao Brightfield-to-Fluorescence Mask Model: Partial Training Report

OrganoidAgent

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1 Status

The previous tmux training run did not finish. It stopped after epoch 20 because a PyTorch DataLoader worker segfaulted:

```
RuntimeError: DataLoader worker (pid 1116658) is killed by signal: Segmentation fault.  
YICHAO_SEGMENTATION_FAILED 2026-05-11T09:05:03+08:00 status=1
```

Therefore this is not a final early-stopped model report. There is no final `test_metrics.json` produced by the normal training script. To still quantify the run, the saved best checkpoint was evaluated separately with `num_workers=0`, which avoids the worker crash.

2 Training Setup

The run trained a mask-only model:

$$B \rightarrow \hat{M}$$

where B is the brightfield organoid crop and M is a cleaned binary fluorescence positive mask derived from the fluorescence channel. The model used the original, stricter mask targets, not the later relaxed masks.

Item	Value
Run root	<code>analysis-outputs/yichao.fluorescence.segmentation/runs/global_gated_unet_v1</code>
Target root	<code>analysis-outputs/yichao.fluorescence.segmentation</code>
Input size	384×384
Batch size	8
Model	Global-gated U-Net with ASPP context
Training split size	6691 crops
Validation split size	1601 crops
Test split size	1088 crops
Best checkpoint	epoch 6
Stopped epoch	20, failed by DataLoader worker segfault

3 Quantification

The best validation checkpoint was epoch 6. Validation improved early, but the run did not complete and validation metrics became unstable afterward.

Metric	Validation, best checkpoint	Test, best checkpoint
Best threshold	0.300	0.600
Best precision	0.274	0.064
Best recall	0.194	0.066
Best F1	0.227	0.065
Best F0.5	0.253	0.065
Best IoU	0.128	0.034
Pixel AP	0.159	0.034
Global AUROC	0.756	0.813
Global AP	0.798	0.688

The interpretation is mixed. The global expression score has signal on the test set (AUROC = 0.813), so the model is learning image-level information. Pixel-level mask prediction is poor on the test set (F0.5 = 0.065, IoU = 0.034), so the model is not reliably localizing expression regions under the original target definition.

4 Training Curves

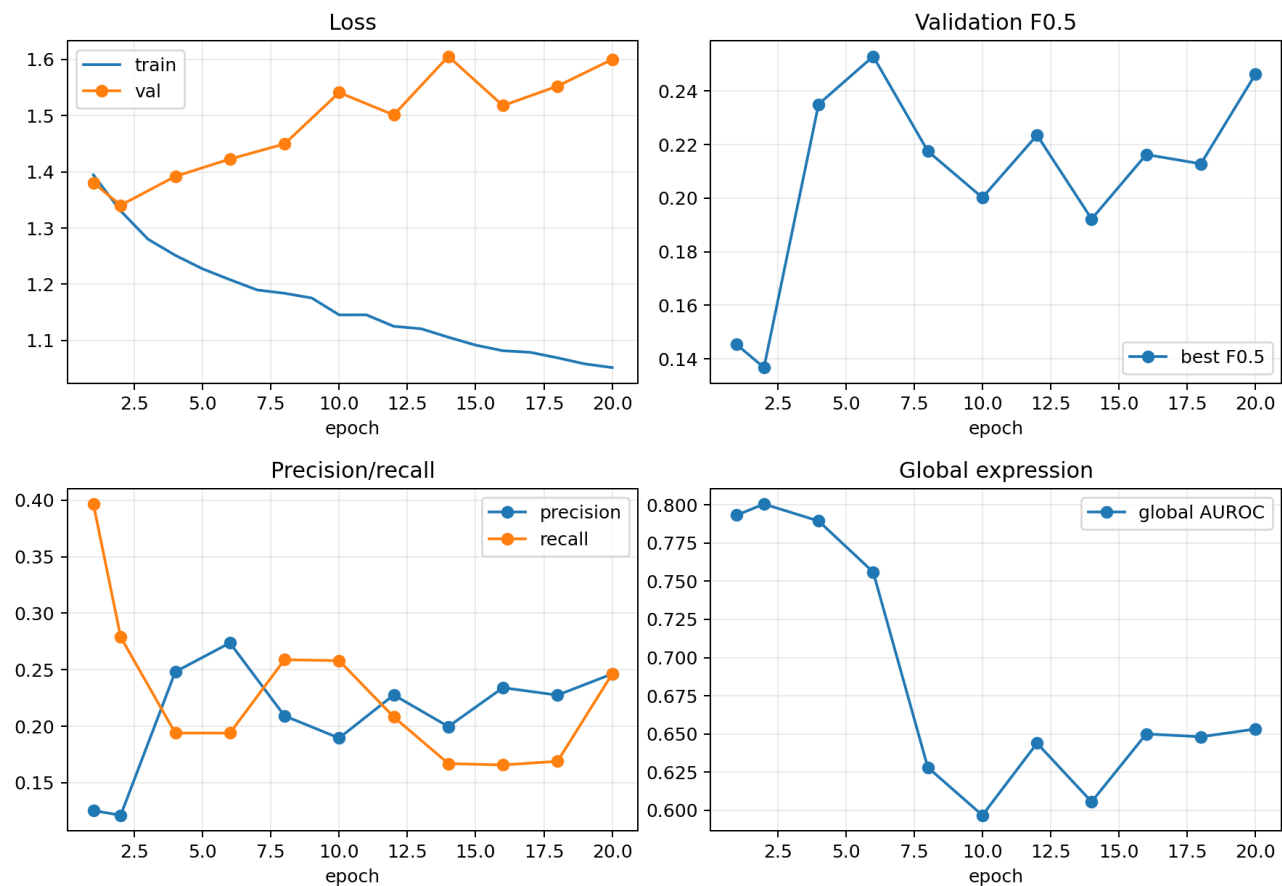


Figure 1: Partial training curves until the run failed. Training loss decreased from 1.394 to 1.052, but validation loss increased and validation metrics were unstable.

5 Prediction Visualization

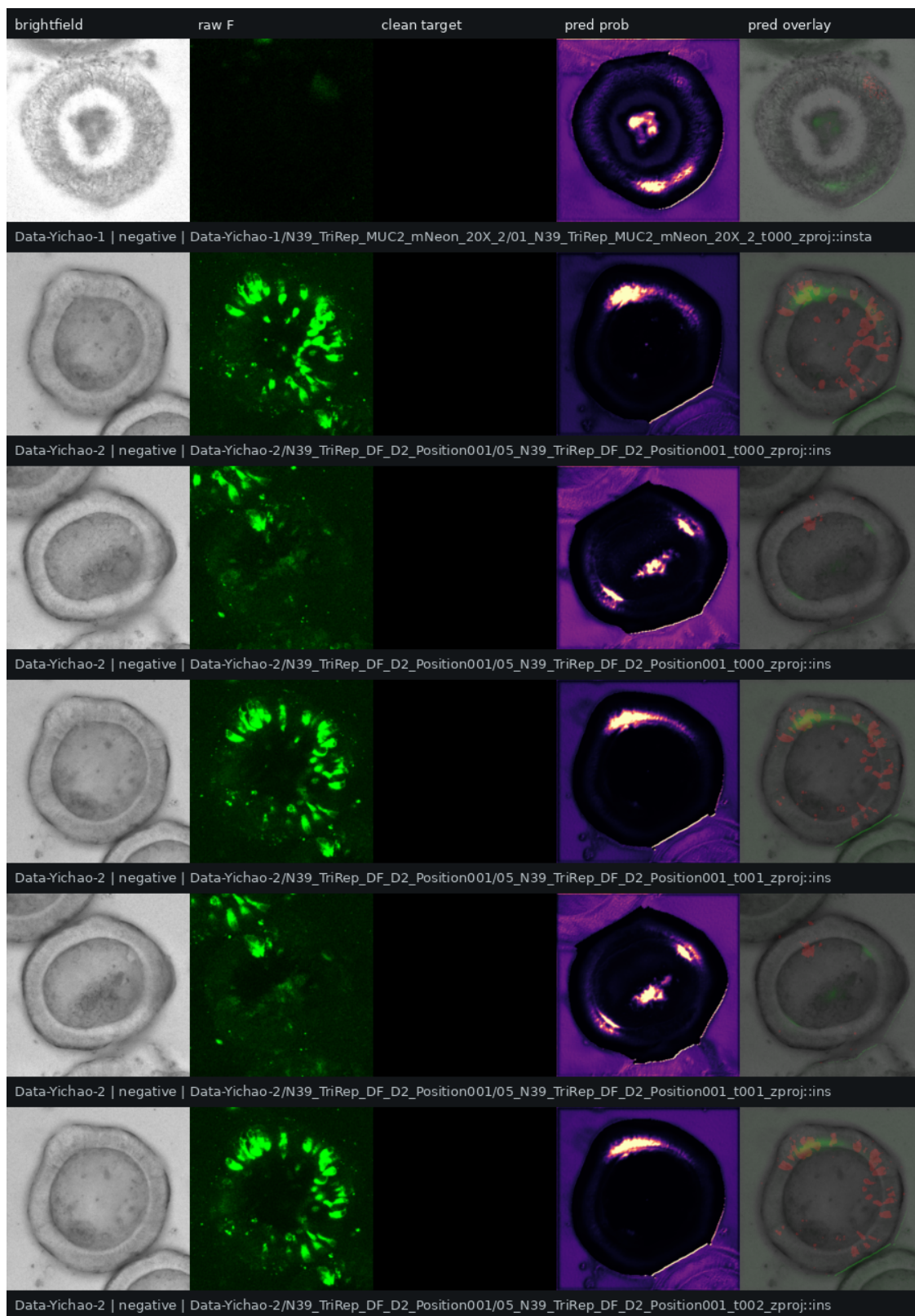


Figure 2: Test-set predictions from the saved best checkpoint. Columns show brightfield, raw fluorescence, clean target, predicted probability, and overlay. Several fluorescence positive looking examples are labeled as negative/blank under the original strict target, which makes this run hard to interpret as a biology-valid model.

6 Target Quality

The original target masks were intentionally conservative. After visual inspection, they appear too aggressive for this dataset and miss many real fluorescence-positive regions. This explains why the model can learn a global signal but struggles to learn reliable pixel-level localization.

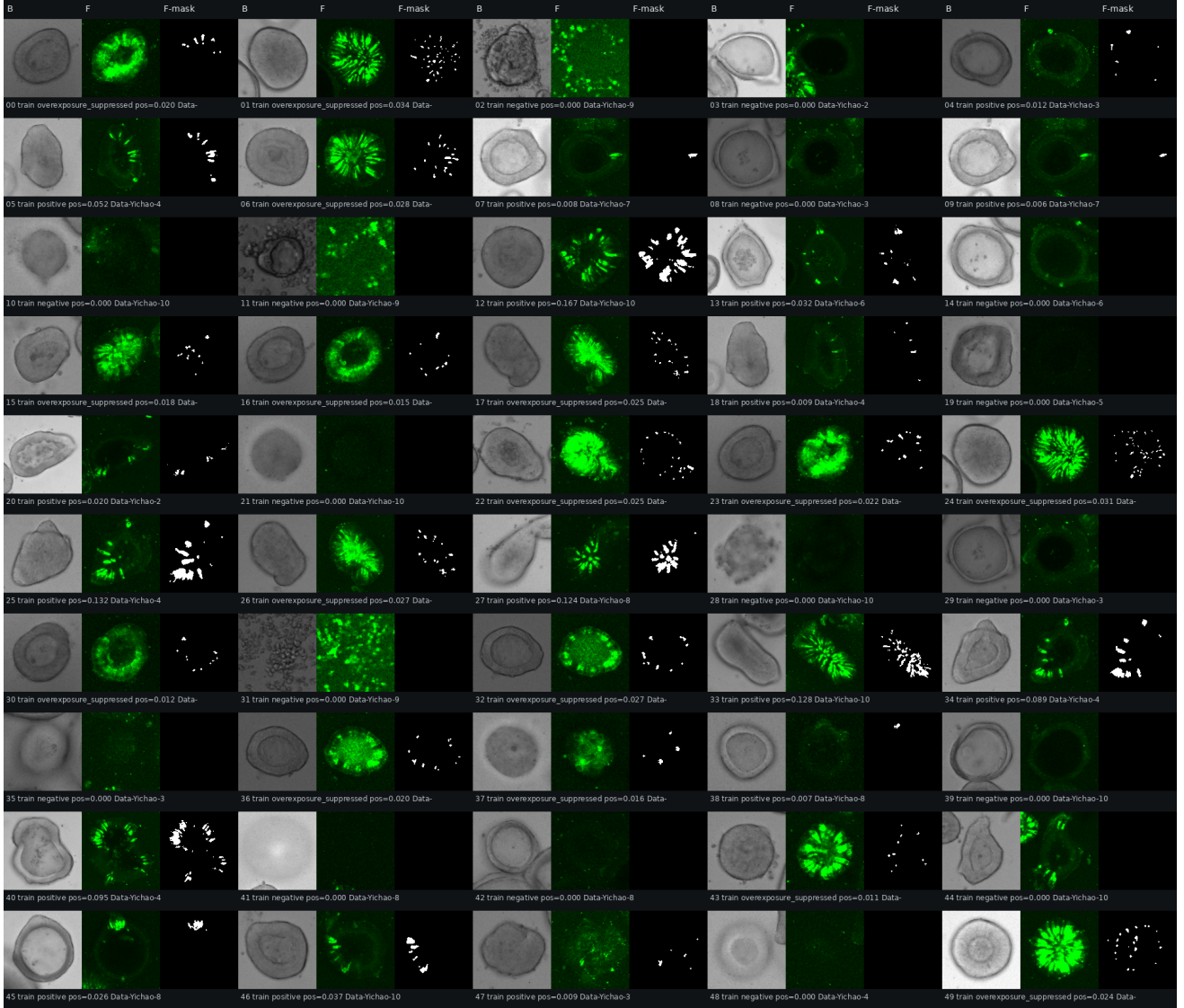


Figure 3: Original strict clean masks used by the failed run. The masks are sparse and often miss weak or thin fluorescence signal.

A relaxed target generation preset was then added. It lowers the robust fluorescence thresholds, disables the morphological opening step that removed thin signal, preserves smaller objects, and keeps full-field overexposure suppression. This relaxed version should be used for the next training run.

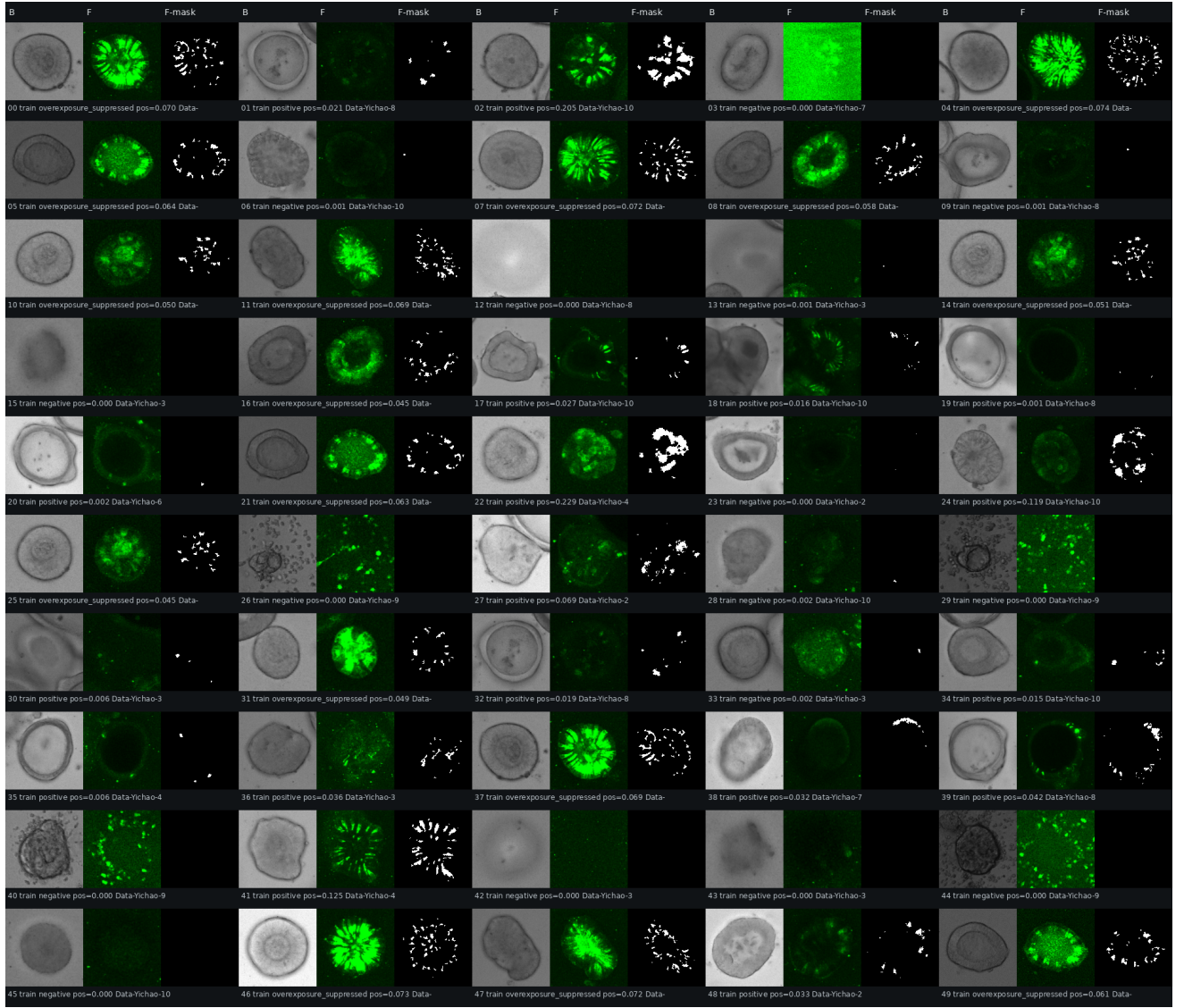


Figure 4: Relaxed B/F/mask examples. These targets recover more weak and peripheral signal while still suppressing many obvious blank or debris-heavy examples.

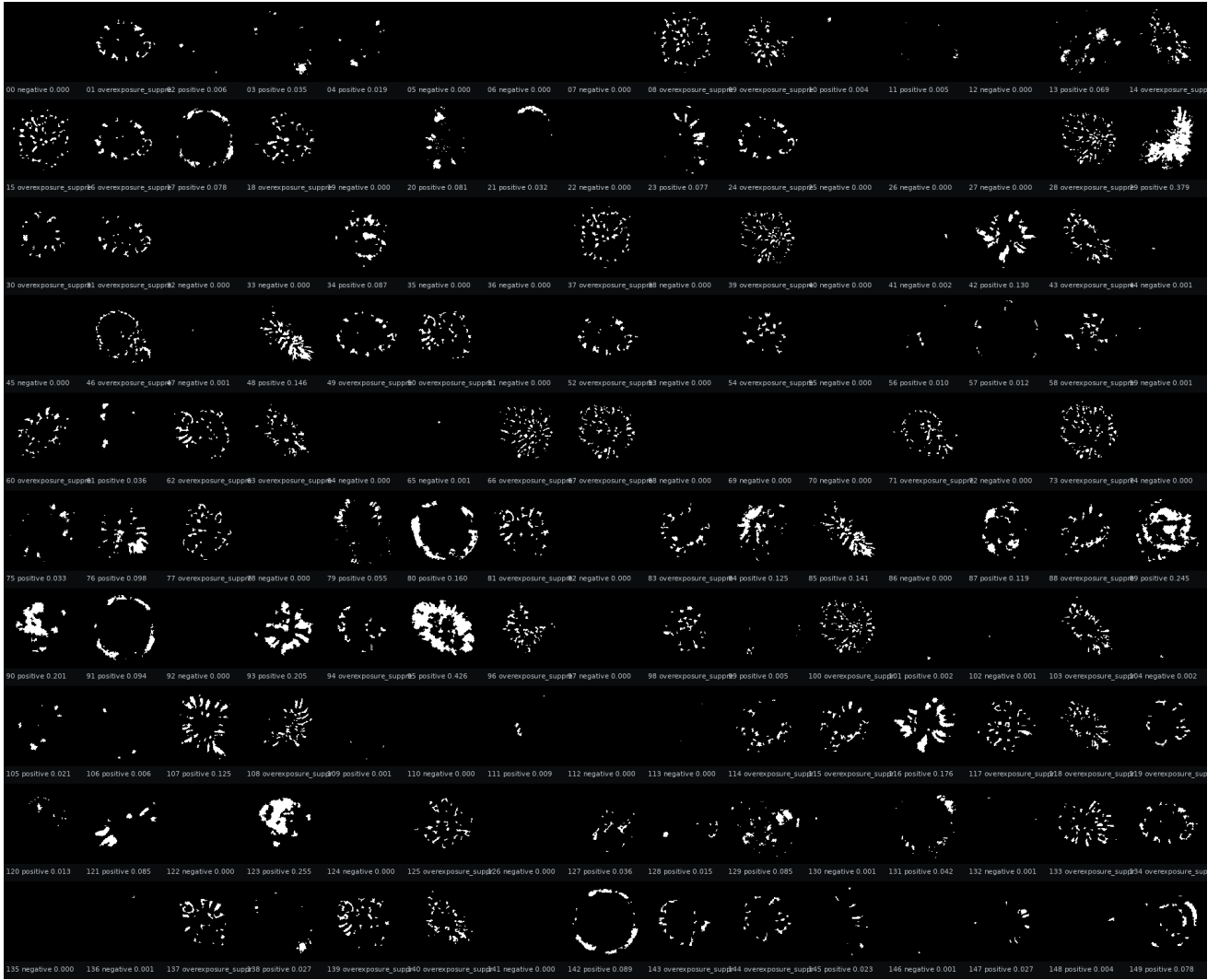


Figure 5: Pure relaxed positive masks. White pixels are the final positive mask. Black pixels are not positive.

7 Explanation

The run is best understood as a failed baseline rather than a final result:

1. The model did learn during training; training loss decreased steadily.
2. Validation mask metrics improved early, with best validation F0.5 at epoch 6.
3. The test pixel-level mask result was weak, so localization did not generalize.
4. The test global AUROC was substantially better than random, so the brightfield still contains expression-related information at the organoid level.
5. The old target masks were too conservative and sometimes marked real fluorescence as negative, which creates contradictory supervision.

6. The run failed because a multi-worker DataLoader process segfaulted, not because the model reached early stopping.

8 Recommended Next Step

Do not treat this checkpoint as the final model. The next run should:

1. Use the relaxed target root: `analysis-outputs/yichao_fluorescence_segmentation_relaxed`.
2. Run with `num_workers=0` or a safer worker configuration to avoid the segfault.
3. Use the three-level model design:

$$B \rightarrow \hat{G}, \quad B \rightarrow \hat{M}, \quad B \rightarrow \hat{A}, \quad \hat{F} = \hat{M}\hat{A}.$$

4. Select checkpoints by both region metrics and image-level expression metrics.